SET 17: EMPLOYEE PERFORMANCE ANALYSIS

Aim

To analyze employee performance using R programming by creating a line chart, bar chart, table, and interactive dashboard.

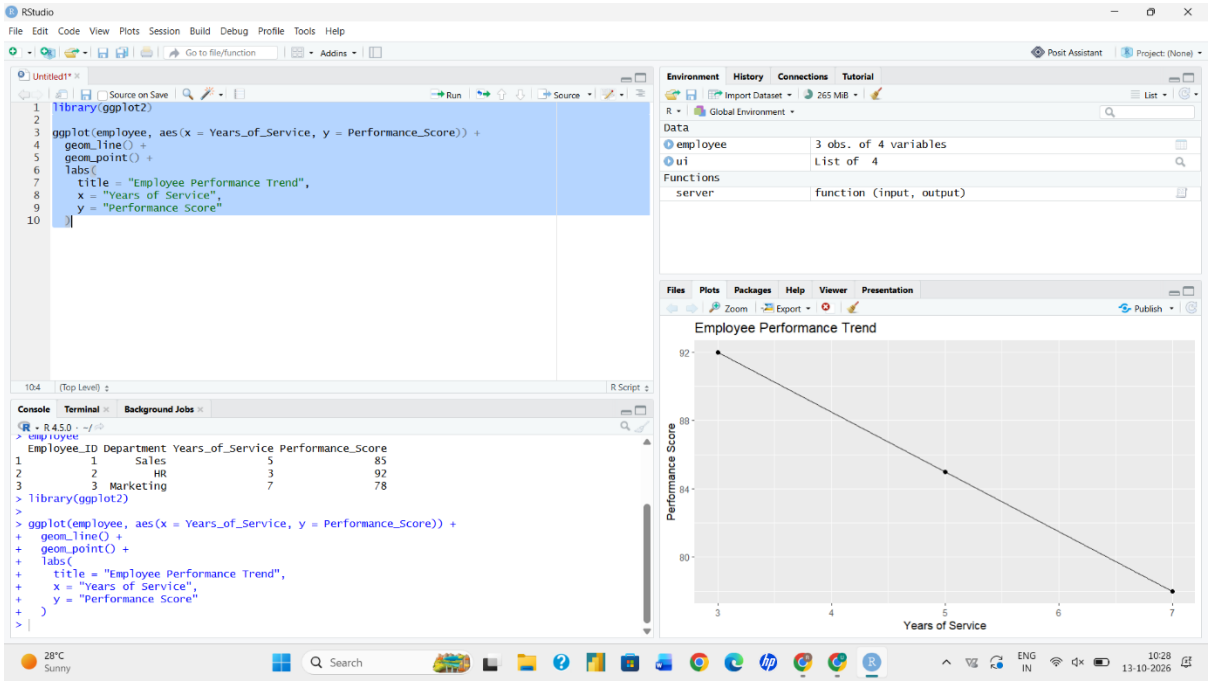
Dataset

The dataset contains Employee ID, Department, Years of Service, and Performance Score.

R Code – Create Dataset

```
employee <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)
```

```
employee
```



Question 1: Create a Line Chart for Employee Performance

R Code

```
library(ggplot2)
```

```
ggplot(employee, aes(
  x = Years_of_Service,
  y = Performance_Score
)) +
  geom_line() +
  geom_point() +
  labs(
    title = "Employee Performance Trend",
    x = "Years of Service",
    y = "Performance Score"
  )
```

Result

The line chart shows employee performance according to their years of service. The highest performance score is 92.

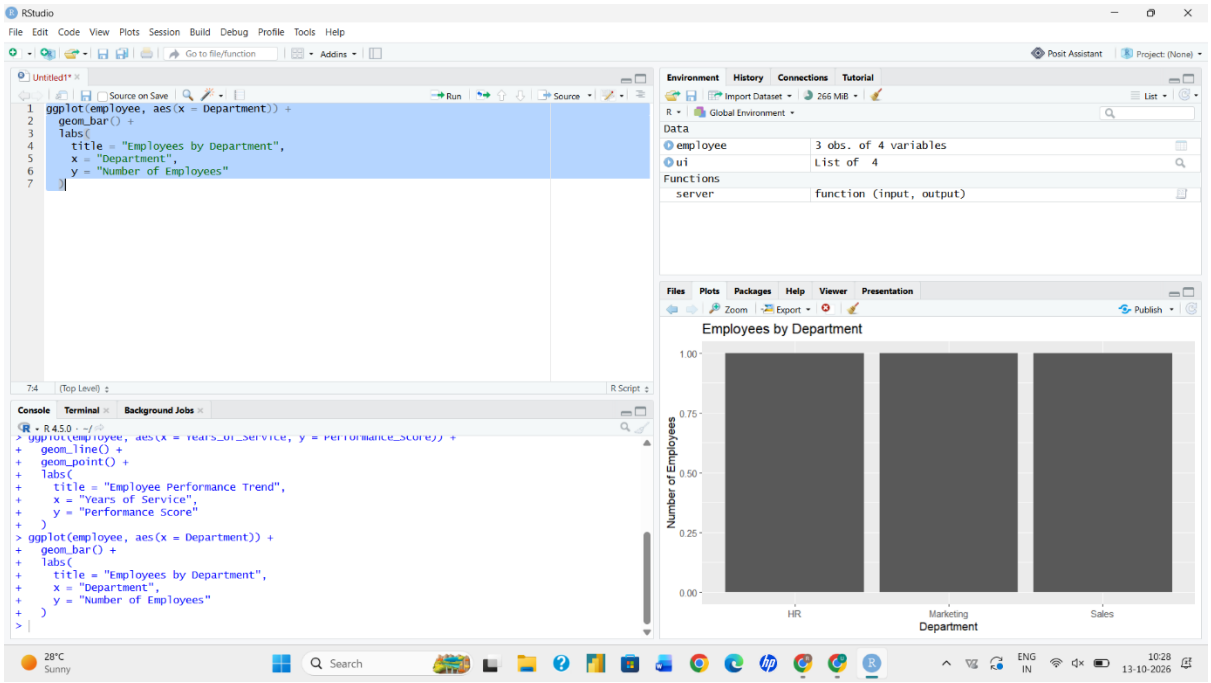
Question 2: Generate a Bar Chart for Departments

R Code

```
ggplot(employee, aes(x = Department)) +
  geom_bar() +
  labs(
    title = "Employees by Department",
    x = "Department",
    y = "Number of Employees"
  )
```

Result

The bar chart shows the number of employees in Sales, HR, and Marketing departments.



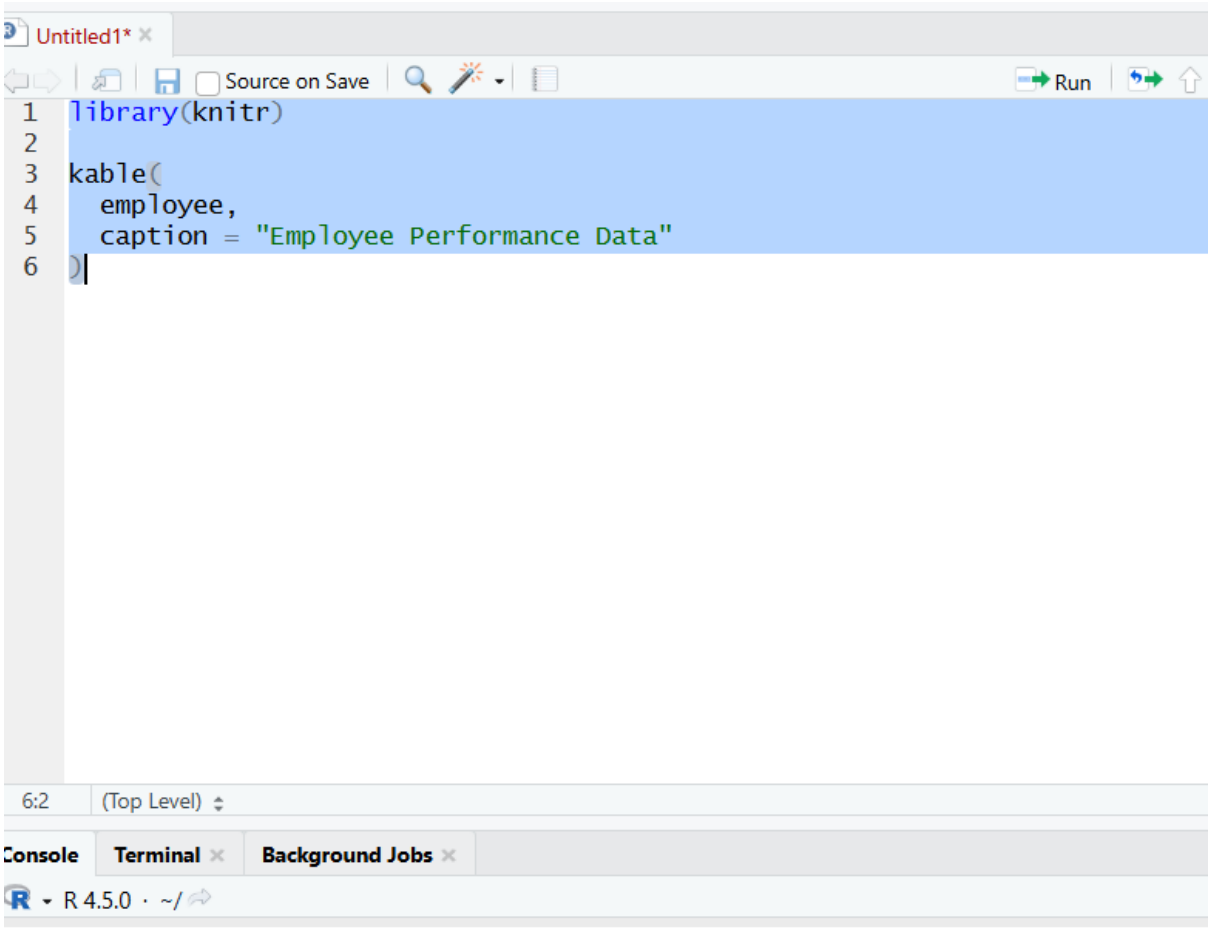
Question 3: Build a Table for Employee Performance

R Code

employee

Result

The table displays Employee ID, Department, Years of Service, and Performance Score.



Question 4: Create an Interactive Dashboard

R Code

library(shiny)

library(ggplot2)

library(plotly)

library(DT)

```
ui <- fluidPage(
  titlePanel("Employee Performance Dashboard"),

  plotlyOutput("performancePlot"),
  plotlyOutput("departmentPlot"),
  DTOutput("employeeTable")
)

server <- function(input, output) {

  output$performancePlot <- renderPlotly({
    p <- ggplot(
      employee,
      aes(x = Years_of_Service, y = Performance_Score)
    ) +
      geom_line() +
      geom_point() +
      labs(
        title = "Employee Performance Trend",
        x = "Years of Service",
        y = "Performance Score"
      )

    ggplotly(p)
  })

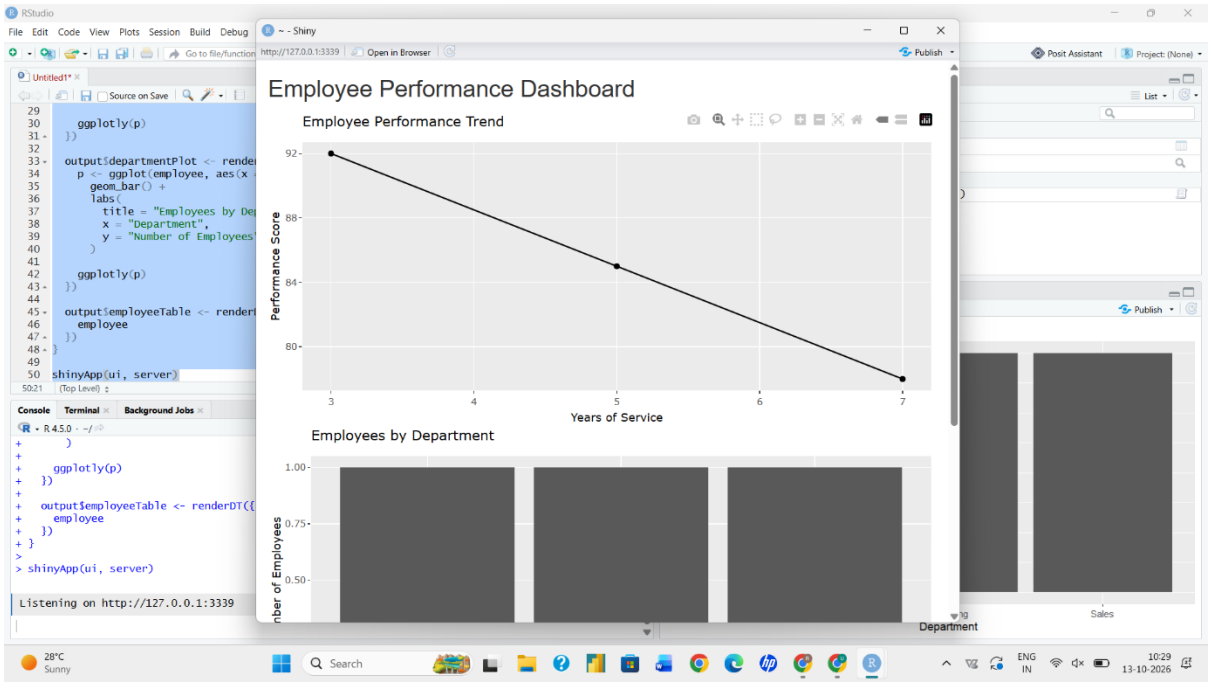
  output$departmentPlot <- renderPlotly({
    p <- ggplot(employee, aes(x = Department)) +
      geom_bar() +
```

```
    labs(
      title = "Employees by Department",
      x = "Department",
      y = "Number of Employees"
    )

  ggplotly(p)
 })

 output$employeeTable <- renderDT({
  employee
 })
}

shinyApp(ui, server)
```



Result

An interactive employee performance dashboard is created containing the line chart, department bar chart, and employee table.

Conclusion

Employee performance data was successfully analyzed using R. Charts, tables, and an interactive dashboard were created to understand employee performance and department information.